

Learning Note: Editing Record

2 October 2026

Purpose

Record substantive changes between editing sessions. This is a dated human-readable record, without numbered releases, approval procedures, or duplicate versions. Future edits append a new dated entry and regenerate this PDF.

2 October 2026 — Initial note

Created: `Electromagnetism-Foundations/main.tex` and its PDF, plus independent learning-plan and editing-record documents in `Documents/`.

Content: Operational definitions and experimental access to electric and magnetic fields; magnetic moments, electron spin, stationary magnets, and energy; definitions of flux, divergence, and curl; fluid, heat, current, and electromagnetic applications; illustrative calculations; and a limited map of related physical laws.

Clarifications from the supplied discussion:

- Electric field is external force per signed test charge in a specified frame, with negligible probe disturbance.
- The magnetic force is perpendicular to velocity and field; the field need not be perpendicular to velocity. One velocity direction cannot determine every field component.
- Spin is intrinsic angular momentum, not literal mechanical rotation. Atomic beam splitting is distinguished from a free-electron experiment.
- Fixed-dipole torque, force, and interaction-energy formulas state their approximation and distinguish uniform from nonuniform external fields.
- Static magnetic fields and intrinsic spin do not require continuous power; resistive coils have losses.
- Field flux is not necessarily material transport. Zero divergence does not mean zero field. Circular field lines do not necessarily mean nonzero local curl.
- Definitions and mathematical theorems are distinguished from physical laws. The history is not reduced to a universal definitions-first sequence.

Build and organization: Self-contained pdfLaTeX source with embedded references and diagram, suitable for Overleaf. Stable filenames, one topic folder, and only TeX/PDF document files intended for Git. Build artifacts are excluded locally.

Validation: Compiled locally with pdfLaTeX, resolved citations on a second pass, and inspected rendered PDF pages. Recompilation does not require external images, BibTeX, or shell escape.

Next: Begin Maxwell's equations only when requested, following the learning plan. Wave propagation, reflection, and detailed AN720 analysis remain future lessons.